Cloud Help Desk Simulation Lab

Lab Documentation & Project Report

Azure Entra ID | ServiceNow PDI | GitHub

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1. Project Overview

This report documents the design, implementation, and outcomes of a cloud-based IT help desk simulation lab built using Microsoft Azure Entra ID and ServiceNow Personal Developer Instance (PDI). The lab was developed as a portfolio project to demonstrate core IT support competencies without requiring local virtual machine infrastructure.

1.1 Objectives

- Simulate a real-world enterprise IT environment using free cloud platforms
- Demonstrate identity and access management (IAM) skills using Azure Entra ID
- Configure and operate an ITSM ticketing system using ServiceNow PDI
- Produce professional documentation suitable for a GitHub portfolio
- Build practical experience directly applicable to entry-level IT support roles

1.2 Tools & Platforms

Platform	Purpose
Microsoft Azure Entra ID	Cloud identity and access management (IAM)
ServiceNow PDI (Zurich)	IT service management (ITSM) and ticketing
GitHub	Portfolio documentation and version control

1.3 Fictional Company Profile

Company Name	CarterTech Inc
Environment	Cloud-based Microsoft Entra ID tenant
Departments	IT Department, HR Department
Total Users	8 provisioned accounts
IT Staff Group	Marcus Webb, Priya Nair, Daniel Ortega, Robert Chen (IT Manager)
HR Staff Group	Sandra Kim, James Tran, Aaliyah Foster, Lisa Morales (HR Director)

1.4 Skills Demonstrated

- User account provisioning and lifecycle management
- Security group creation and membership management
- Multi-Factor Authentication (MFA) enforcement
- Account lockout, disable, and reinstatement procedures
- Self-Service Password Reset (SSPR) policy research and design
- Conditional Access policy design and licensing awareness
- Audit log review and administrative accountability
- ITSM ticket creation, documentation, and resolution
- SLA definition and priority-based response targets
- Knowledge Base article creation for self-service support
- Escalation procedures and Tier 2 handoff documentation

2. Azure Entra ID — Identity & Access Management

Microsoft Azure Entra ID (formerly Azure Active Directory) is Microsoft's cloud-based identity and access management platform. It serves as the backbone of modern enterprise IT environments, replacing traditional on-premises Active Directory for organizations moving to cloud infrastructure.

2.1 Tenant Setup

An Azure free account was created at portal.azure.com using a personal Microsoft account to ensure full administrative control. The tenant was configured under the name CarterTech Inc with a custom onmicrosoft.com domain.

Tenant Name	CarterTech Inc
Account Type	Azure Free Tier ($200 credit)
Authentication	Personal Microsoft account (gmail-linked)
Instance Created	March 29, 2026

2.2 Task 1 — User Account Provisioning

Eight user accounts were provisioned across two departments to simulate a small enterprise environment. Each account was configured with a consistent username format (firstname.lastname@tenant.onmicrosoft.com), job title, and department assignment.

User	Department
Marcus Webb	IT Department
Priya Nair	IT Department
Daniel Ortega	IT Department
Robert Chen	IT Manager
Sandra Kim	HR Department
James Tran	HR Department
Aaliyah Foster	HR Department
Lisa Morales	HR Director

> *Note: All 8 accounts confirmed active in Entra ID Users list. Screenshot captured showing all accounts with correct department assignments.*

2.3 Task 2 — Security Group Configuration

Two security groups were created to organize users by department and enable group-based access control. Security groups are the foundation of role-based access management in enterprise environments.

Group Name	IT Staff
Type	Security
Source	Cloud
Members	Marcus Webb, Priya Nair, Daniel Ortega, Robert Chen

Total Members	4
Created	March 29, 2026 at 10:18 PM

Group Name	HR Staff
Type	Security
Source	Cloud
Members	Sandra Kim, James Tran, Aaliyah Foster, Lisa Morales
Total Members	4
Created	March 29, 2026 at 10:19 PM

2.4 Task 3 — Multi-Factor Authentication (MFA)

MFA enforcement was configured using Per-User MFA, the free-tier alternative to Conditional Access policy-based MFA enforcement. All 8 CarterTech user accounts were enabled for MFA through the Per-User MFA console in the Entra ID Users panel.

> *Limitation: Conditional Access policy-based MFA requires Entra ID P1 or P2 licensing. The free tier supports Per-User MFA only. Conditional Access policy was designed and documented but could not be applied without a premium license. This is a known real-world licensing constraint that IT administrators regularly encounter.*

Conditional Access policy designed (not applied due to licensing):

- Policy name: Require MFA for All Users
- Scope: All users, All cloud apps
- Control: Grant access — Require multifactor authentication
- Mode: Would be set to Report-only initially, then On after testing

2.5 Task 4 — Location-Based Access Policy

A location-based Conditional Access policy was researched and designed to block sign-ins from outside the United States. Implementation requires Entra ID P1 licensing.

> *Limitation: Same Entra ID P1 licensing requirement as Task 3. Policy design documented as proof of knowledge.*

- Policy name: Block Sign-in Outside US
- Named location: Approved US Location (marked as trusted)
- Scope: All users
- Control: Block access for any location except Approved US Location

2.6 Task 5 — Password Reset (Account Lockout Simulation)

A password reset was performed on Marcus Webb's account to simulate the most common helpdesk ticket type — account lockout. This procedure was completed entirely within the Azure portal.

Target User	Marcus Webb
Action Performed	Password reset via Entra ID admin console
Setting Applied	Must change password on next sign-in
Temporary Password	[Redacted for documentation]
Outcome	Password reset successful — confirmed in audit log

Date	March 30, 2026 at 9:57 PM

2.7 Task 6 — Account Disable & Reinstatement (Offboarding Simulation)

Daniel Ortega's account was disabled to simulate an employee offboarding procedure, then re-enabled to simulate account reinstatement. In a real environment, disabling an account is the first action taken when an employee is terminated.

Target User	Daniel Ortega
Action 1	Account disabled — Block sign-in enabled (10:06 PM)
Account Status	Disabled — confirmed in profile overview
Action 2	Account re-enabled — Block sign-in removed (10:07 PM)
Final Status	Enabled — confirmed with success toast notification
Outcome	Full offboarding/reinstatement cycle documented

2.8 Task 7 — Self-Service Password Reset (SSPR)

SSPR was researched and the configuration was fully planned. Implementation requires Entra ID P1 licensing on the free tier.

> *Limitation: SSPR requires Entra ID Premium licensing. Configuration was designed and documented as follows.*

- Scope: IT-Staff security group (scoped rollout best practice)
- Authentication methods required: 1
- Available methods: Email, Mobile phone
- Registration enforcement: Users prompted on next login
- Self-service portal: aka.ms/sspr

2.9 Task 8 — Audit Log Review

The Entra ID audit log was reviewed to verify all administrative actions taken during the lab. Audit logs provide an accountability trail that is essential for compliance and security incident investigation.

The following actions were confirmed in the audit log:

- Enable account — Daniel Ortega — Success (10:07 PM, March 30)
- Disable account — Daniel Ortega — Success (10:06 PM, March 30)
- Update user — Daniel Ortega — Success (10:06 PM, March 30)
- Reset password — Marcus Webb — confirmed via separate password reset log
- Create user — All 8 accounts — confirmed via provisioning logs

> *Note: Audit log screenshot captured showing UserManagement category entries with Success status. All actions attributed to the tenant admin account.*

3. ServiceNow PDI — IT Service Management

ServiceNow is the industry-leading IT Service Management (ITSM) platform used by the majority of enterprise IT organizations. A Personal Developer Instance (PDI) was provisioned through the ServiceNow Developer Program at developer.servicenow.com — a fully featured, free instance running the Zurich release.

3.1 Instance Details

Instance ID	dev356444
Instance URL	dev356444.service-now.com
Release Version	Zurich (latest available)
User Role	Admin
Provisioned	March 31, 2026

3.2 Task 1 — SLA Definitions

Three Service Level Agreement definitions were created to establish response and resolution time targets for incidents by priority. SLAs are critical to helpdesk operations as they define performance expectations and drive escalation decisions.

SLA Name	Priority
P1 - Critical	1 Hour resolution — 24 hour schedule
P2 - High	4 Hours resolution — 24 hour schedule
P3 - Moderate	8 Hours resolution — Business hours schedule

> *Note: All three SLAs configured on the Incident [incident] table. Screenshot captured showing all three definitions in the SLA Definitions list alongside existing system SLAs.*

3.3 Task 2 — User Profiles

Three CarterTech user profiles were created in ServiceNow to serve as callers and requesters on incident tickets.

- Marcus Webb — IT Department
- Sandra Kim — HR Department
- Robert Chen — IT Manager

3.4 Task 3 — Incident Tickets

Five realistic incident tickets were created and worked through the full ITIL incident management lifecycle. Each ticket documents the issue, troubleshooting steps taken, and resolution — mirroring real helpdesk documentation standards.

INC001 — User cannot log in - account locked out

Ticket Number	INC001
Caller	Marcus Webb (Abel Tuter — demo user)
Priority	2 - High

Category	Inquiry / Help
State	Resolved
Short Description	User cannot log in - account locked out
Description	User reports seeing account locked message when attempting to log in to their workstation. Multiple failed login attempts detected.
Work Notes	1. Verified user identity by confirming employee ID and department. 2. Checked Entra ID audit logs — confirmed 5 failed login attempts in the past 10 minutes. 3. Reset user password in Azure Entra ID with must change on next login flag enabled. 4. Communicated temporary password to user via verified phone callback. 5. Confirmed user was able to log in successfully and set new password.
Resolution Notes	Account was locked due to multiple failed login attempts. Password reset performed in Entra ID. User confirmed successful login. Advised user to enable SSPR for future self-service resets.

INC002 — MFA not working after getting new phone

Ticket Number	INC002
Caller	Sandra Kim (demo user)
Priority	2 - High
Category	User Account
State	Resolved
Short Description	MFA not working after getting new phone
Description	User received a new phone and Microsoft Authenticator is no longer generating valid codes. User cannot access any company applications requiring MFA.
Work Notes	1. Verified user identity. 2. Confirmed user has a new device and old Authenticator data was not transferred. 3. Navigated to Entra ID > Users > Sandra Kim > Authentication methods. 4. Deleted old, registered authenticator device. 5. Generated Temporary Access Pass for user to re-register MFA. 6. Guided user through re-registering Authenticator on new device. 7. Confirmed user can access applications successfully.
Resolution Notes	MFA registration was tied to previous device. Old authentication method removed and Temporary Access Pass issued. User successfully re-registered Authenticator on new phone. Advised user to set up a backup authentication method.

INC003 — New HR employee needs access to HR shared resources

Ticket Number	INC003
Caller	James Tran (demo user)
Priority	3 - Moderate
Category	Request
State	Resolved
Short Description	New HR employee needs access to HR shared resources

Description	New HR employee requires access to HR-Staff shared resources. Currently has no group membership and cannot access department files.
Work Notes	1. Verified request came from HR Director Lisa Morales via email confirmation. 2. Navigated to Entra ID > Groups > HR-Staff. 3. Added James Tran as a member of the HR-Staff security group. 4. Confirmed group membership propagated after 5 minutes. 5. Followed up with user to confirm access granted successfully.
Resolution Notes	Access request approved by HR Director. User added to HR-Staff security group in Entra ID. Access to HR shared resources confirmed active.

INC004 — IT Manager cannot connect to company VPN from home

Ticket Number	INC004
Caller	Robert Chen (demo user)
Priority	2 - High
Category	Network
State	In Progress — Escalated to Tier 2
Short Description	IT Manager cannot connect to company VPN from home
Description	IT Manager is unable to connect to company VPN from home office. Receiving Authentication failed error. On-site access works normally.
Work Notes	1. Verified user identity and confirmed issue is remote VPN only, on-site access works. 2. Checked Entra ID sign-in logs, no failed authentication events found for this user. 3. Confirmed VPN client version is up to date. 4. Attempted basic troubleshooting: restarted VPN client, cleared credentials cache, issue persists. 5. Issue is beyond Tier 1 scope — requires network team access to VPN gateway logs. 6. Escalating ticket to Tier 2 Network team with full troubleshooting notes attached.
Resolution Notes	Tier 1 troubleshooting completed. VPN client verified current, credentials cache cleared, Entra ID sign-in logs checked with no errors found. Issue escalated to Tier 2 Network team for VPN gateway investigation. Ticket remains open and assigned to Network queue.

INC005 — New hire account setup request — Aaliyah Foster

Ticket Number	INC005
Caller	Lisa Morales (demo user)
Priority	3 - Moderate
Category	Request
State	Resolved
Short Description	New hire account setup request — Aaliyah Foster
Description	New HR employee starting Monday requires full account setup including Entra ID account creation, security group membership, and welcome email with login instructions.
Work Notes	1. Received new hire details from HR Director: Name: Aaliyah Foster, Title: HR Coordinator, Start date: Monday.

	2. Created Entra ID user account: aaliyah.foster@cartertech.onmicrosoft.com. 3. Set temporary password with must change on first login enabled. 4. Added user to HR-Staff security group. 5. Filled in all profile fields: job title, department, manager set to Lisa Morales. 6. Sent welcome email template to HR Director with login credentials and SSPR setup instructions.
Resolution Notes	New hire account fully provisioned in Entra ID. User added to HR-Staff security group. Temporary credentials delivered to HR Director for onboarding. Account ready for Monday start date.

3.5 Task 4 — Knowledge Base Article

A Knowledge Base article was authored and submitted for publication to enable self-service password reset without helpdesk intervention. Knowledge Base articles are a key ticket deflection strategy in enterprise helpdesk environments.

Article Number	KB0010001
Knowledge Base	IT
Category	IT
Title	How to Reset Your Password Using Self-Service Password Reset (SSPR)
State	Scheduled for publication (April 2, 2026)
Audience	All CarterTech employees

Article content summary: Step-by-step instructions for users to reset their own password via aka.ms/sspr, including identity verification via email or mobile phone, and escalation path to helpdesk at ext. 100 if SSPR registration is not complete.

4. Key Findings & Lessons Learned

4.1 Licensing Constraints

Several planned tasks required Entra ID P1 or P2 premium licensing, which is not available on the free tier. These included Conditional Access policies and Self-Service Password Reset configuration. This is a realistic and important finding — in production environments, IT administrators regularly navigate licensing decisions that affect which security controls can be implemented.

Rather than treating this as a gap, the experience of identifying the licensing boundary, understanding why features are gated, and documenting the policy design regardless of implementation is itself a valuable skill demonstrated in this lab.

4.2 Cloud vs. On-Premises

All tasks in this lab were completed without any local virtual machine infrastructure, demonstrating that modern IT support skills can be developed and demonstrated entirely through cloud platforms. Azure Entra ID replicates the identity management functions of on-premises Active Directory, while ServiceNow PDI provides a full enterprise ITSM environment in the browser.

4.3 ITIL Alignment

All five incident tickets were documented following the ITIL incident management lifecycle: identification, logging, categorization, prioritization, diagnosis, resolution, and closure. The escalation ticket (INC004) demonstrates understanding of Tier 1 vs. Tier 2 boundaries and proper handoff procedures.

4.4 Security Awareness

Throughout the lab, several security best practices were applied and documented: MFA enforcement for all accounts, forced password change on reset, immediate account disablement on simulated offboarding, audit log review for accountability, and scope-limited SSPR rollout (IT-Staff group only for initial deployment).

5. Project Completion Summary

5.1 Azure Entra ID — Completed Tasks

- ✓ Azure free account created and tenant configured as CarterTech Inc
- ✓ 8 user accounts provisioned across IT and HR departments
- ✓ IT-Staff and HR-Staff security groups created with correct membership
- ✓ Per-User MFA enabled for all accounts (free tier)
- ✓ Conditional Access MFA policy designed and documented (premium-gated)
- ✓ Location-based access policy designed and documented (premium-gated)
- ✓ Password reset performed for Marcus Webb with forced change on login
- ✓ Account disabled and re-enabled for Daniel Ortega (offboarding simulation)
- ✓ SSPR policy researched and documented (premium-gated)
- ✓ Audit logs reviewed and administrative actions verified

5.2 ServiceNow PDI — Completed Tasks

- ✓ PDI provisioned on Zurich release at developer.servicenow.com
- ✓ P1, P2, and P3 SLA definitions created on Incident table
- ✓ Three CarterTech user profiles created
- ✓ INC001 — Account lockout — Created and resolved
- ✓ INC002 — MFA new phone — Created and resolved
- ✓ INC003 — Access request — Created and resolved
- ✓ INC004 — VPN escalation — Created and escalated to Tier 2
- ✓ INC005 — New hire onboarding — Created and resolved
- ✓ Knowledge Base article KB0010001 authored and submitted for publication

5.3 Resume Bullets

The following bullets are ready for use on a resume under a Lab Experience section:

Cloud Help Desk Simulation Lab — Azure Entra ID / ServiceNow PDI / GitHub

- Provisioned and managed a simulated enterprise identity environment in Azure Entra ID; created 8 user accounts across two departments, configured security groups, and enforced MFA for all users.
- Simulated common IT support scenarios including account lockouts, password resets, user offboarding/reinstatement, and new hire provisioning — all documented in Azure Entra ID audit logs.
- Built and resolved 5 end-to-end incident tickets in ServiceNow PDI spanning account lockouts, MFA failures, access requests, new hire onboarding, and a Tier 2 escalation; documented each ticket following ITIL incident management lifecycle.
- Configured ServiceNow SLA definitions (P1–P3) and authored a Knowledge Base article for self-service password reset, demonstrating ITSM operations and ticket deflection strategy.
- Identified and documented Entra ID P1 licensing constraints affecting Conditional Access and SSPR implementation — demonstrating real-world awareness of enterprise licensing and security policy tradeoffs.

6. GitHub Repository Structure

The following structure should be used when publishing this project to GitHub. All screenshot files should be organized by phase and named descriptively.

```
cloud-helpdesk-lab/
├── README.md
├── lab-report.docx  (this document)
├── azure-entra-id/
│   ├── entra-id-setup.md
│   └── screenshots/
├── servicenow/
│   ├── sn-setup.md
│   └── screenshots/
├── incident-tickets/
│   ├── INC001-account-lockout.md
│   ├── INC002-mfa-new-phone.md
│   ├── INC003-access-request.md
│   ├── INC004-vpn-escalation.md
│   └── INC005-new-hire-onboarding.md
└── runbook/
    └── helpdesk-runbook.md
```

> *Note: Upload all screenshots taken during the lab into the appropriate screenshots/ folders. Name them descriptively, e.g. 01-users-list-all-8-accounts.png, 02-it-staff-group-4-members.png, etc.*